Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/24/2005
PI Name:	Hollingsworth, Mark D. Ph.D.		
Project Title:	Crystal Growth of New Families of Ferroelastic Materials		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Electronic materials		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mdholl@ksu.edu	**Fax:**	FY 785-532-6666
PI Organization Type:	UNIVERSITY	**Phone:**	785-532-2727
Organization Name:	Kansas State University		
PI Address 1:	111 Willard Hall		
PI Address 2:	Department of Chemistry		
PI Web Page:			
City:	Manhattan	**State:**	KS
Zip Code:	66506	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	06/01/2000	**End Date:**	11/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Carpenter, Paul	**Contact Phone:**	'256 544-5149
Contact Email:	Paul.Carpenter@msfc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG8-1702		
Performance Goal No.:			
Performance Goal Text:			

This research program seeks to understand the cooperative elastic phenomenon of domain switching through comparison of physical properties and defect structures of a series of closely related ferroelastic inclusion compounds. These inclusion compounds are based upon the same urea channel framework, but they differ according to the orientation, dimensionality and cooperativity of specific hydrogen bonding networks. By growing crystals in the presence of tailor-made impurities, one can disrupt certain hydrogen bond networks and observe measurable changes in the kinetics and reversibility of the ferroelastic response. This approach therefore serves as a powerful method for understanding the specific molecular mechanisms that control domain switching.
Twinning and dislocations are shown to have a profound impact on the domain switching properties of these materials, so a significant fraction of this work will be devoted to refining the protocols for growth of large crystals containing a minimum of dislocations and twins. These will serve as a series of controls that will ultimately be compared with crystals grown under microgravity conditions.

Task Description:

Evaluation of the defect structures of crystals grown on the ground and in space will be conducted with synchrotron white beam X-ray topography. These topographic studies will be conducted on crystals at ambient pressure, but also on crystals under applied anisotropic stresses. In tandem with these measurements, domain switching processes will be studied with photographic and videographic techniques. This will allow a direct correlation between defect structures and the domain switching responses of crystals grown under different conditions and will add significantly to our understanding of the domain switching process. Our research is in the area of solid state organic chemistry, which lies at the interface between physical organic chemistry and materials science. We use crystalline solids as models to ask and answer fundamental questions about physical processes, molecular interactions and chemical reactions that are important for fabrication, stabilization and application of technological materials. Much of our most recent work has focused on the phenomena of ferroelastic and ferroelectric domain switching, in which application of an external force or electric field to a crystal causes the molecules inside the crystal to reorient, in tandem, to a new, orientational state.

To better understand and control the domain switching process, we have designed and synthesized approximately twenty closely related, ferroelastic organic crystals. (This almost doubles the number of known organic ferroelectrics and ferroelastics.) Our approach has been to use crystalline inclusion compounds, in which one molecule (the guest) is trapped or "included" within the crystalline framework of a second molecule (the host). By keeping the host constant and varying the proportions and kinds of guests, it has been possible to tailor these materials so that domain switching is rapid and reversible (which is desirable for high technology applications).

Unfortunately, any material that undergoes domain switching so easily can also make mistakes during crystal growth. Most of the crystals that we grow in the laboratory contain numerous domains, defects and dislocations, all of which complicate both the measurement and interpretation of domain switching phenomena. Because we have direct evidence for the link between solution turbulence and defect formation, crystal growth in the microgravity of space offers the best method for growing defect-free crystals. In the anticipation of future space-based crystal growth experiments, our efforts in this grant are focused on refining the protocols for crystal growth of these materials and on using a variety of techniques to measure the domain switching properties of crystals that we grow in the laboratory. Our research is in the area of solid state organic chemistry, which lies at the interface between physical organic chemistry and materials science. We use crystalline solids as models to ask and answer fundamental questions about physical processes, molecular interactions and chemical reactions that are important for fabrication, stabilization and application of technological materials. Much of our most recent work has focused on the phenomena of ferroelastic and ferroelectric domain switching, in which application of an external force or electric field to a crystal causes the molecules inside the crystal to reorient, in tandem, to a new, orientational state. Because the orientations of molecules in a crystal influence the way the crystal interacts with light, domain switching processes are important for optical switching and other high technology applications. By studying the elastic properties of crystals that undergo domain switching processes, we hope to obtain a better understanding of how to design new materials that can undergo domain switching at very high rates.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress - Summary
Contents 1. Large-scale organic synthesis of guests for ferroelastic inclusion compounds 2. Search for polymorphism in 2,10-undecanedione/urea 3. Mixed inclusion compounds of Br(CH2)6CN and Cl(CH2)6CN in urea 4. Absolute configuration and chiral twinning in UICs 5. Brillouin and Raman scattering of ferroelastic urea inclusion compounds 6. Static versus dynamic surface roughening during crystal growth of UICs 7. Ultrafast videomicroscopy of domain wall motions 8. Crystal structures of 1,11-undecanedioic acid/urea and related systems 9. Chiral twin boundaries and inhibition of domain wall motion 10. Polar order, NLO effects and piezoelectric effects in mixed inclusion crystals 11. NLO-active inclusion compounds of thiourea containing mixtures of guests 12. Electric field poling of a urea inclusion compound 13. Mechanism of rubber-like behavior and memory effects in ferroelastic UICs

1. Large-scale organic synthesis of guests for ferroelastic inclusion compounds

One of our first goals was to design an efficient and inexpensive synthesis of a series of bis(methylketone)s that serve as guests for our studies of ferroelastic domain switching in urea inclusion compounds. Graduate student Terry Geiger conducted these studies. Previously, we had prepared 2,10-undecanedione by reacting azeloyl chloride with lithium dimethyl cuprate. This expensive procedure was reliable, but it could only be run on a 1-2 gram scale, and the product required purification by column chromatography. We therefore experimented with several methods, including the following: (1) Reaction of 1,7-dicyanoheptane with an excess of either methyl lithium or methyl magnesium bromide. The imine that is formed should be readily converted into 2,10-undecanedione by hydrolysis, but this reaction was plagued by solubility problems (of both the monoanion and the dianion) and by acid-base reactions, which gave side products. This method was therefore abandoned. (2) Formation of the bis-Grignard reagent from 1,7-dibromoheptane, reaction of this with two equivalents of acetaldehyde and bleach oxidation of the diol to 2,10-undecanedione. Once again, the reaction of the Grignard reagent with the aldehyde gave several side products under a variety of conditions, so this procedure was not pursued further. (3) Acetoacetic ester synthesis. In this reaction, 1,5-dibromopentane is reacted with an excess of the enolate of ethyl acetoacetate. Saponification of the diester followed by acidic decarboxylation gives 2,10-undecanedione in one pot. After significant refinement of conditions, we have made this reaction work with remarkable success. Through the use of a twofold excess of enolate, we have avoided the formation of a possible side product (acetyl cyclohexane) produced by ring closure of the initially formed enolate with the pendant alkyl bromide. Under optimal conditions, this reaction gives a 92% yield of a product that can be purified by running it through a plug of silica gel and recrystallization. The starting materials are inexpensive, and the reaction can be done on a large scale. The high purity of this guest or possibly absence of specific impurities has resulted in several single-domained crystals of 2,10-undecanedione/urea, which we have seen only rarely in several years of crystal growth of these materials. We have synthesized more than 100 grams of 2,10-undecanedione and have used a large fraction of this material in this research program. This acetoacetic ester synthesis has also been used to synthesize over 20 grams of 2,15-hexadecanedione, which is a guest that forms ferroelastic inclusion compounds with urea. Previously, we have had no more than 500 mg of this guest, and we were not able to conduct systematic studies of crystal growth of the urea inclusion compound. (We had enough to solve the crystal structure and conduct a limited number of stress studies, but we have never had enough of this material for systematic studies of crystal growth.) We modified this synthetic method

to prepare 2,19-eicosanedione, which is ferroelastic with a commensurate relationship of 2cg' = 5ch'. These crystals diffracted poorly, however, so it is not clear that we can solve this structure.

2. Search for polymorphism in 2,10-undecanedione/urea

Our "helical wheel" method has allowed us to predict the topology of host-guest hydrogen bonding in a variety of urea inclusion compounds with bis(methylketone)s. (See Chem. Mater., 8, 1588-1591 (1996).) For certain guests, including 2,10-undecanedione, this formalism suggested an alternative, polymorphic packing mode in which the guest adopts a slightly different conformation in the channel. (In the observed polymorph, the guest undergoes a small counterclockwise torsional twist in a right-handed channel. In the alternative arrangement, the guest undergoes a small clockwise torsional twist in the same right-handed channel. Since the (very small) torsion angles are essentially the same, but in opposite directions, these arrangements are essentially diastereomeric.) Because the ferroelastic distortion of the two modifications should be collinear, the second polymorph should be indistinguishable from the observed modification with optical microscopy. Since the second polymorph could have a different ferroelastic response from the known one, we decided to search for this second form by growing 2,10-undecanedione/urea crystals under a variety of conditions and then solving the structures of these crystals using X-ray diffraction. In the end, Dr. Matthew Peterson solved the structures of four crystals of 2,10-undecanedione/urea grown from different batches and found that each of them had the same structure as the known polymorph. Comparison with analogous bis(methylketone) structures in a homologous series of UICs showed that in every case, the "sense" of the guest torsion (clockwise or counterclockwise) was opposite that of the urea host structure, which forms either right or left handed helices. Using the Cerius2 molecular modeling package, comparison of the known structures with those of the diastereomeric (simulated) structures convinced us that host-guest hydrogen bonding only occurs with ureas that turn into the "major groove" of this double helix structure. This can only occur if the sense of the guest torsion is opposite that of the urea host because rotation of ureas in the opposite direction would give rise to unfavorable host-host interactions. Given the uniform adherence of all of the different inclusion compounds (both within and between samples with different chain lengths), we conclude that it is very unlikely that we will be able to form the second polymorph. Proving the non-existence of the second polymorph is difficult, but its existence would be inconsistent with our current understanding of this inclusion system.

3. Mixed inclusion compounds of Br(CH2)6CN and Cl(CH2)6CN in urea

We have identified a number of biaxial UICs in which the host molecules adopt non-helical hydrogen bonding arrangements. With the exception of one C2/c structure, these centrosymmetric (P21/n) channel structures are best described as stacked loops of urea hexamers. This alternative packing mode is routinely adopted for guests with the formula X(CH2)6Y (X or Y = -Br, -Cl, -CN, -NC) when the host:guest stoichiometry is 6:1. These materials exhibit distortions of up to 10% from hexagonal metric symmetry, but most do not exhibit ferroelastic domain reorientation even though they are potentially ferroelastic. For these non-helical systems, a simple rotation-translation motion of the guest would produce an inequivalent structure; ferroelastic (i.e. degenerate) domain reorientation would require reorientation of the n-glide symmetry element by approximately 60°, but this would necessitate a large-scale reorganization of the urea framework and concomitant rupture of numerous hydrogen bonds. In effect, continuous motion of the guest down the channel is disrupted by the non-helical nature of these stacked loop structures. Depending upon the orientation of the distortion, these stacked loop structures may be again separated into two classes, ones that exhibit either [100] distortion and ones that exhibit [001] distortion of the host framework. Because of the preferred orientation of the guests in the channels, UICs formed from Br(CH2)6Br, Br(CH2)6Cl and Cl(CH2)6Cl are distorted along [100] by as much as 10% from hexagonal metric symmetry. In contrast, UICs of NC(CH2)6CN and CN(CH2)6NC exhibit somewhat smaller distortions of approximately 2% along the [001] axis. Although UICs of NC(CH2)6CN and CN(CH2)6NC are again potentially ferroelastic, the kinetic barriers to domain reorientation in these stacked loop systems are apparently too high for observation of this phenomenon. Since chlorine and bromine give rise to one sort of distortion and cyano gives rise to the other, we decided to investigate the physical and structural properties of Cl(CH2)6CN/urea and Br(CH2)6CN/urea as well as "solid solutions" containing mixtures of Cl(CH2)6CN and Br(CH2)6CN. Both Cl(CH2)6CN/urea and Br(CH2)6CN/urea have high temperature ferroelastic phases in which the distortion from hexagonal symmetry is less than one percent. Both of them also distort to a low temperature (non-ferroelastic) phase that is isomorphic with Br(CH2)6Br/urea and its congeners. For Cl(CH2)6CN/urea, the phase transition is at –68°C, whereas for Br(CH2)6CN/urea it is at +20°C. Differential scanning calorimetry studies done by Dr. Brian Dinkelmeyer shows that the phase transition temperatures for "solid solutions" of (Cl(CH2)6CN + Br(CH2)6CN) in urea exhibit a linear dependence on mole fraction of guest. For certain "solid solutions" in this series, the differential scanning calorimetry traces indicated more than one phase transition between room temperature and –68°C. In collaboration with Prof. Michael Dudley SUNY, Stony Brook), we therefore investigated the possibility of intrasector zoning of guests in these materials. SWBXT clearly shows some form of growth banding with in given sectors of these mixed inclusion compounds. For this unstirred crystal growth medium, this is consistent with a cooperative process in which preferred deposition of Cl(CH2)6CN (for example) and epitaxial constraints enrich the sample in that guest until the surrounding solution becomes enriched in Br(CH2)6CN. Further crystallization generates a growth band enriched in Br(CH2)6CN, which is concomitantly depleted in the surrounding solution. Enrichment of Cl(CH2)6CN and Br(CH2)6CN therefore alternates on a macroscopic scale that is related to the growth rate of the crystal, diffusion rates of guests in solution, and factors affecting the epitaxial matching of the domains enriched in either of the guests. Extensive X-ray diffraction studies of UICs of NC(CH2)6CN, CN(CH2)6NC, Cl(CH2)6CN, and Br(CH2)6CN by Drs. Kevin Pate, Matthew Peterson and Brian Dinkelmeyer have provided several insights into the links between guest conformations, lattice strain and domain switching processes in these UICs. By refining the occupancies of guest conformers at different temperatures, we have arrived at a fairly self-consistent picture for the relationship between conformer population and lattice strain in these materials. Thus far, it appears that a critical ratio of conformer populations is required for a ferroelastic phase transition to occur in NC(CH2)6CN/urea, CN(CH2)6NC/urea. In Cl(CH2)6CN/urea and Br(CH2)6CN/urea, however, this phase transition appears to be driven by steric constraints, and it is remarkable since the guest molecules must undergo translations of 5.5Å down the channel axis. Such large-scale motions are unexpected, and they are especially provocative in the present case because alternative pathways exist for relaxation of guest-induced strain.

4. Absolute configuration and chiral twinning in UICs

As part of our effort to understand domain switching in UICs, we had earlier used optical microscopy and SWBXT to study chiral (Brasil) twinning in certain uniaxial UICs. This form of twinning is counterintutive, but its existence is readily discernible with optical microscopy in several uniaxial UICs, such as 2,12-tridecanedione/urea. Unfortunately, the biaxial optical properties of ferroelastic UICs makes it impossible to observe Brasil twinning, so we decided to use SWBXT to assess the amount of strain at the Brasil twin boundaries. Quite surprisingly, SWBXT shows that these

Brasil twin boundaries are almost strain-free, so this makes it difficult to know if chiral twinning exists in 2,10-undecanedione/urea. Comparison of the chiral twin boundary with the structures of centrosymmetric UICs (see below) allowed us to develop a molecular mechanism for the Brasil twinning in helical UICs. Because the glide relation at this Brasil twin boundary gives rise to a high barrier to domain switching, we suspect that it is not important in the ferroelastic UICs where we have observed domain wall motion along both {110} and {130} type boundaries. As part of this study, we have attempted to use the Bijvoet method to determine the absolute configuration of dextrorotatory and levorotatory domains of 2,12-tridecanedione/urea. Since this crystal contains no heavy atoms, Dr. Matthew Peterson has synthesized nine different guests containing heavy atoms (Br, Cl, I, Hg) that were expected to give trigonal UICs with flat plate habits, well-defined commensurate structures and uniaxial optical properties. In each case, however, the crystal was either centrosymmetric, biaxial or had a complex superstructure. Although there may be other possible guests that satisfy the optical and crystallographic requirements, neutron diffraction of a known system that is stereospecifically labeled with deuterium ((5S,6S)-2,9-decanedione-d2/urea) is now the most viable option. (Bijvoet analysis with X-ray diffraction of unlabeled material is infeasible since the crystal contains only C, H, N and O.) Our proposal to the Intense Pulsed Neutron Source at Argonne National Lab. has been well-received by our collaborator Thomas Koetzle, who is an expert on determination of absolute structures from neutron diffraction. We have worked out most of the steps of the synthesis of this chiral deuterium labeled guest, but we are out of funds for this project and will not be able to complete it until we receive more funds. This synthetic project has been carried forward in an advanced undergraduate laboratory, where the students worked out the conditions for two of the six steps in this synthesis.

Task Progress:

Security StatementPrivacy StatementDisclaimer StatementAccessibility StatementContact UsResponsible NPRS Official: Task Book help desk The low temperature crystal structure of 2-decanone/urea is quite similar to that of 2,9-decanedione/urea, but careful inspection of the difference maps indicated extra carbonyl oxygen electron density shifted up and down the channel by 1.8-1.9Å from the original carbonyl poge in crystal habit (from plates to needles) when the mole fraction of 2-decanone was increased above ~50%. We had been confused because these needles exhibited diffuse X?ray scattering indicative of one-dimensional ordering of guests, but the layer line spacings indicated that they were commensurate structures (which we thought should have givels of 2,9-decanedione/urea, SWBXT is the only method that can distinguish domains related by the six-fold screw twinning, but we were surprised to find that mixed inclusion compounds of 2,9-decanedione and 2?decanone in urea are optically anomalous. That is, they have trigonal metric symmetry bu even with small amounts of 2-undecanone, the crystals form needles instead of plates. Urea inclusion compounds containing mixtures of 2,9-decanedione and 2-decanone exhibit variation in crystal habit that depends on the guest composition. In our earlier work on these mixed crystals, we had found that there was ring crystal growth, the guests are locked into position because they form hydrogen bonded polymers within a channel. But nearby channels can be offset by one urea helix unit and still have essentially the same free energy. The hydrogen bonded polymers are correlated between channels only over short distances, so the diffraction is almost absent for two out in many ways this is a better system than mixed crystals containing 2,9-decanedione and congeners. Unfortunately, crystals containing low concentrations of 2,10-undecanedione exhibit no clearly-defined rotational twin boundaries. Under stress, these crystals exhibit broad, diffuse domain walls that are characteristic of materials that have small distortions from high symmetry. Crystals containing somewhat larger amounts of 2,10-undecanedione are probably better suited for these studies, but we need to find samples in which the linear birefringence is weak enough for us to simultaneously observe chiral twinning. Further experimentation should reveal which crystals have the proper optical characteristics for observation of this phenomenon.

10. Polar order, NLO effects and piezoelectric effects in mixed inclusion crystals

Following the NLO work on the optically anomalous crystals, it became apparent that our ferroelastic crystals of 2,10-undecanedione/urea might also incorporate impurities in a polar fashion. The major focus of this grant has been to elucidate the role that impurities play in changing the ferroelastic domain switching from a plastic (irreversible) process into an elastic (reversible) one, so we have studied numerous mixed crystals. We were extremely pleased to find that our ferroelastic inclusion compounds containing mixtures of guests show strong NLO activity that is again dependent on the orientation of the crystal in the laser. Although we are still trying to prove it with X-ray diffraction (of a 60:40 crystal containing 2,10-undecanedione and 2-undecanone), we expect that the monoketone guest preferentially enters the crystal in a polar fashion (with its carbonyl pointing into the growth face of the crystal). The mixed crystal that we have been studying with X-ray diffraction has space group P1 (instead of orthorhombic C2221 for the pure 2,10-undecanedione/urea), so we expect that the orientational disorder will occur in a polar fashion with the polar axis having a large component in the plane perpendicular to the channel axis. It is difficult to imagine how such strong NLO activity would occur in any other way. We have recently solved the crystal structure of a mixed UIC containing 75% 2,10-undecanedione and 25% 10-bromo-2-decanone, which is a heavy atom analog of 2-undecanone. In space group P1, this structure is tractable, and it is possible to refine the occupancies of the minor guest in four different sites. However, the small differences in the refined populations suggest that with this minor guest, anyway, the molecular recognition processes leading to lowered symmetry are quite subtle. Although the differences in site occupancy factors were small in this one particular case, other systems, including those containing larger amounts of monoketone, show more striking effects. By using a variety of approaches, we have made great progress toward understanding the molecular recognition events that control the polar ordering of guests in these inclusion compounds. In 2002 we proposed a crystal growth model that identifies the different types of guest binding sites that are accessible on the different growth faces of an alkanedione/urea crystal. For a trigonal crystal such as 2,9-decanedione/urea were predicted that every other growth face of the crystal will expose equivalent binding sites for the guests, as long as the chirality of the crystal remains constant and the sense of the screw-dislocation responsible for growth remains the same throughout the crystal growth process. By mapping the birefringence of inclusion compounds containing 2,9-decanedione and other minor guests (e.g. 2-decanone), we have shown that, in many cases, the retardation (a measure of the concentration and ordering of minor guest) alternates between different growth sectors. In crystals that exhibit chiral twinning (which can be observed in the microscope), the guest recognition is reversed in an understandable way. Careful analysis reveals that even in highly textured crystals, it is possible to formulate a self-consistent mechanism for minor guest incorporation in which a single screw dislocation is responsible for the growth.

11. NLO-active inclusion compounds of thiourea containing mixtures of guests

Thiourea also forms inclusion compounds, but the channels are large enough to accommodate aromatic guests and the hydrogen bond topology is not helical. In these crystals, the channel forms an hourglass structure in which a guest the size of benzene can fit within one "lobe" of the hourglass. Many of the thiourea inclusion compounds are commensurate structures with rhombohedral symmetry, whereas others are distorted and have monoclinic symmetry. We have prepared

several thiourea inclusion compounds containing azobenzene and analogous compounds, and we have determined the crystal structures of several of these materials, which exhibit large distortions from trigonal metric symmetry. These distortions are too large for observation of ferroelastic domain switching, but our work on mixed crystals convinced us that introduction of a polar impurity molecule into a high symmetry thiourea crystal would produce a new series of piezoelectric and ferroelectric inclusion compounds. Because the distortion from trigonal symmetry should be either vanishingly small or absent, and because the guests are not cooperatively bound together in large arrays, ferroelectric domain switching should be much easier than in the urea inclusion compounds. We have shown that thiourea inclusion crystals grown from a 9:1 mixture of chlorocyclohexane and pyridine N-oxide exhibit second harmonic generation. The chlorocyclohexane/thiourea structure is a centrosymmetric structure with rhombohedral symmetry (R3bar space group), so the introduction of the polar guest is implicated in the NLO response. Because these crystals grow as long needles, we have not yet looked at the domain structures, which will require us to cut and polish a thin crystal. These crystals probably grow by the same mechanism that we have proposed for urea inclusion compounds, so we should be able to tailor the crystal habit by using a more polar solvent. Currently, we are growing crystals containing other mixtures of guests. We believe that the polar incorporation of impurities into high symmetry inclusion compound structures is an important new way to generate large families of ferroelectric and piezoelectric materials. Because these materials are also NLO active, this becomes a new way of generating switchable NLO materials.

12. Electric field poling of a urea inclusion compound

In the course of our work, it has become clear that the barrier to domain switching can be reduced if the ferroelastic distortion away from high symmetry is as small as possible. We had shown that incorporation of more than 16% of 2-undecanone into crystals of 2,10-undecanedione/urea lowers the barrier to domain switching enough to make this process reversible. The complex optical textures observed for crystals containing large amounts of 2-undecanone had discouraged us from pursuing crystal growth from mixtures containing mostly 2-undecanone, but we viewed these as good candidates for ferroelectric domain switching since the distortion was small and the guests were highly mobile (from solid state 2H NMR). Nevertheless, UIC crystals grown from a 4:1 mixture of 2-undecanone and 2,10-undecanedione are optically anomalous and exhibit very clear growth-face dependent incorporation and orientation of 2,10-undecanedione. We have not yet observed electric field-induced domain switching with crystals of this composition, but we have succeeded with crystals grown from a 94:6 mixture of 2-undecanone and 2,10-undecanedione. These crystals are again optically anomalous, but they exhibit a multitude of sectors that show no simple correlations with growth faces. When a relatively small electric field (5 kV/cm) is applied to this crystal a large fraction of the crystal becomes aligned with the slow axis of the optical indicatrix approximately 30° from the axis of the electric field. In many ways, this is a milestone for our research program, since this project was initiated with the hope that we could find ferroelectric behavior in urea inclusion compounds. This poling or electric field induced detwinning provided an immediate opportunity for determining the crystal structure of 2-undecanone/urea (which was previously inaccessible due to extensive twinning), so we excised a fragment that showed uniform birefringence and collected X-ray data sets at room temperature and at -140°C. At room temperature, this inclusion compound is incommensurate (as is 2-undecanone/urea) with 2cg' = 32.5Å (and 3cg' = 33.0Å). At -140°C, the crystal is commensurate with 2cg'=3ch' = 33.0Å. In space group C2221, the structure is fairly well behaved, and the guest chains and guest carbonyls are oriented towards the vertices of the channels. There may be another orientation for the guest carbonyl, but the poling evidently oriented the bulk of the crystal into a single arrangement. Because the room temperature crystal is incommensurate, it will be difficult to solve the complete structure. Nevertheless, Patterson projections onto a plane perpendicular to the channel axis are very similar to those observed for the low temperature structure, suggesting that the guest orientation is similar in the high and low temperature phases. The slow axis of the optical indicatrix at room temperature does not match the orientation of the guest chain in the low temperature structure, but this is obviously a complicated phenomenon.

13. Mechanism of rubber-like behavior and memory effects in ferroelastic UICs

Thanks to support from NASA, we have finally unraveled the detailed mechanism for the "rubber-like behavior" and memory effects that we have observed in the ferroelastic UICs that we study. In essence, SWBXT reveals the presence of nanoscopic twins that are epitaxially matched with the predominant parent domain orientation that exists before application of external stress. Under the application of external stress, these nanoscopic twins become epitaxially mismatched with the surrounding daughter domain and therefore raise the energy of the daughter. The mismatch created in this process provides the driving force for spontaneous domain reversion (and therefore for the rubber-like behavior and memory effects that are observed). The nanoscopic twins responsible for this behavior are not large enough to be visible with ordinary light microscopy (or even birefringence mapping) or even to diffract coherently. But they do appear as dynamic diffraction contrast in the SWBXT experiment. Thus, although they are invisible, they play a critical role in the spontaneous domain reversion and in the specific annealing processes caused by relaxive impurities and acoustomechanical vibrations. We now have unambiguous X-ray structural data that demonstrates the presence of a displacive twin caused by relaxation of the epitaxially mismatched daughter generated from the nanoscopic twins by applying stress. Recent work has shown that in metal alloys that exhibit rubber-like behaviour and ferroelectrics that exhibit "aging-induced microstructure memory", the process of aging after cooling through a symmetry-lowering phase transition allows point defects to conform to the crystal symmetry of the parent, thus providing a bias for the observed memory effects. In the UICs that exhibit the same kind of behavior, the "aging" process occurs during crystal growth itself to generate the epitaxially matched twins that are responsible for the memory effects. Although they are different from the ones that operate in certain shape memory materials, the twinning and defect phenomena responsible for the rubber-like behavior and memory effects in UICs may be generally applicable to domain switching in ferroelastic and ferroelectric crystals and to other solid state processes. Recently, we have submitted a manuscript to Nature on this work, and we were pleased to learn that they have sent the manuscript out for external review. This mechanism explains all of the anomalies that we have observed in the last ten years and has relevance to other materials, so we are optimistic about the fate of our manuscript. It also opens up the door to hypothesis-directed research on the phenomenon of "memory effects," and in particular, how specific types of imperfections and crystal growth conditions (including microgravity) affect the observed memory effects.

Bibliography Type: Description: (Last Updated: 06/24/2005)

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Presentation	Hollingsworth, M. D., Peterson, M. L., Rush, J. R., Abel, M. J., Black, A. A., Kesselring, D. A. and Butenhoff, A. G. "From ferroelastics to ferroelectrics" 39th Congress of the Mexican Chemical Society, Merida, Mexico Oct-2004
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Presentation	Rush, J. R., Hollingsworth, M. D. and Abel, M. J. "Polar ordering and electric field-induced domain reorientation in ferroelastic channel inclusion compounds" Midwest Regional ACS Meeting, Manhattan, KS Oct-2004
Presentation	Rush, J. R., Hollingsworth, M. D. and Abel, M. J. "Polar ordering and electric field-induced domain reorientation in ferroelastic channel inclusion compounds" American Crystallographic Association National Meeting Jul-2004
Presentation	Rush, J. R., Hollingsworth, M. D. and Abel, M. J. "Polar ordering and electric field-induced domain reorientation in channel inclusion compounds" Fifteenth Midwest Organic Solid State Chemistry Symposium, Carbondale, IL Jun-2004
Presentation	Hollingsworth, M. D., Rush, J. R., Peterson, M. L., Champion, B. L., Abel, M. J., and Black, A. A. "Symmetry reduction in ferroelastics: a general approach to new ferroelectric materials?" Sixteenth International Conference on the Chemistry of the Organic Solid State, Sydney, Australia Jul-2003